GenAI Quality and Maturity Assessment

Beyond the Demo

SYSTEM **My Awesome genAI system**	TEAM **My awesome team**
DATE **2026-03-08 13:10**	ASSESSOR **Jon Doe**
QUALITY SCORE **48.00**	ACTUAL MATURITY **L1**
REQUIRED MATURITY **L5**	CRITICALITY **Production Critical**

Below required maturity **Needs major quality improvements**

Maturity Level

L1 L2 L3 L4 L5

Current maturity Expected maturity based on business criticality

Characteristic Scores

Utility

37.5

Economy

25.0

Robustness

50.0

Productionizability

50.0

Modifiability

50.0

Comprehensibility

37.5

Responsibility

70.0

Action Summary

CRITICAL ACTIONS

3

IMPORTANT ACTIONS

18

NICE-TO-HAVE ACTIONS

0

Priority Actions

Critical

Fairness (responsibility)

Current gap: small | Blocks level: L1 | Required quality requirement: full

Evidence: Risks documented, no segmented eval report yet.

Action: Evaluate across relevant demographic groups and implement mitigation for harmful disparities.

Responsiveness (utility)

Current gap: large | Blocks level: L1 | Required quality requirement: full

Evidence: P95 is monitored, P99 alerting missing.

Action: Define and track P95/P99 latency continuously and enforce SLO compliance.

Vulnerability (responsibility)

Current gap: small | Blocks level: L1 | Required quality requirement: full

Evidence: Baseline controls in place, no recent red-team exercise.

Action: Add multi-layered security controls including bot detection, red teaming exercises, PII redaction, jailbreak defenses, and security audits.

Important

Accuracy (utility)

Current gap: small | Blocks level: L3 | Required quality requirement: full

Evidence: Golden set exists but only 80 examples.

Action: Expand to a 200+ example golden dataset with automated evals plus periodic human review.

Discoverability (comprehensibility)

Current gap: large | Blocks level: L3 | Required quality requirement: full

Evidence: Registry entry exists but metadata is incomplete.

Action: Register the system with searchable metadata (e.g., model details, prompts, example usage, limitations, ownership).

Effectiveness (utility)

Current gap: large | Blocks level: L3 | Required quality requirement: min

Evidence: No recent A/B run after prompt and model changes.

Action: Run an A/B experiment tied to a primary business KPI and document results.

Operability (modifiability)

Current gap: small | Blocks level: L3 | Required quality requirement: full

Evidence: Rollback available but no canary process.

Action: Add canary rollouts, safe rollback automation, and runtime kill switches.

Understandability (comprehensibility)

Current gap: small | Blocks level: L3 | Required quality requirement: full

Evidence: Partial docs available, runbook not complete.

Action: Fully document the system's architecture (e.g., prompt engineering or finetunine rationale, known limitations, onboarding guides).

Adaptability (robustness)

Current gap: small | Blocks level: L4 | Required quality requirement: full

Evidence: Manual process for KB refresh exists, no automation.

Action: Automate update loops using feedback and evaluation-gated releases.

Explainability (responsibility)

Current gap: small | Blocks level: L4 | Required quality requirement: full

Evidence: Confidence shown internally, no user-facing citations.

Action: Provide user-facing transparency artifacts (citations, tool traces, confidence).

Repeatability (productionizability)

Current gap: small | Blocks level: L4 | Required quality requirement: full

Evidence: Prompt release process partially automated.

Action: Fully automate update/evaluation flow with versioning.

Traceability (comprehensibility)

Current gap: small | Blocks level: L4 | Required quality requirement: full

Evidence: Prompt versions logged, full lineage not captured.

Action: Capture end-to-end lineage across models, context, tool calls, reasoning traces and post-processing steps.

Cost-Effectiveness (economy)

Current gap: large | Blocks level: L5 | Required quality requirement: full

Evidence: Cost dashboard exists but ROI margin is not tracked.

Action: Step 1: Document and report all major cost components (API costs, storage, compute, human reviews, etc). Step 2: Demonstrate sustained positive ROI and establish budget guardrails.

Efficiency (economy)

Current gap: small | Blocks level: L5 | Required quality requirement: full

Evidence: Prompt updates and evals are partly automated.

Action: Optimize prompts for token efficiency (in case of prompt based model) or fine-tuning for cost-latency-quality efficiency (in case of fine-tuned).

Maintainability (modifiability)

Current gap: small | Blocks level: L5 | Required quality requirement: full

Evidence: Prompts and code versioned, change rationale inconsistent.

Action: Version all key artifacts with structured change rationale.

Modularity (modifiability)

Current gap: small | Blocks level: L5 | Required quality requirement: full

Evidence: Retrieval and post-processing are partially decoupled.

Action: Fully enforce modularity across retrieval, prompting, generation, and safety components.

Monitoring (productionizability)

Current gap: small | Blocks level: L5 | Required quality requirement: full

Evidence: Operational metrics tracked, no continuous quality eval loop.

Action: Add continuous automated output quality evaluation.

Readability (comprehensibility)

Current gap: small | Blocks level: L5 | Required quality requirement: full

Evidence: Prompt style guide followed by most teams.

Action: Enforce unified readability standards for prompts and source code.

Resilience (robustness)

Current gap: small | Blocks level: L5 | Required quality requirement: full

Evidence: Update pipeline success around 84% per quarter.

Action: Raise automated pipeline success rate to at least 90%.

Scalability (robustness)

Current gap: large | Blocks level: L5 | Required quality requirement: full

Evidence: No dynamic scaling policy yet.

Action: Implement dynamic scaling with rate limiting and cost controls.

Testability (modifiability)

Current gap: small | Blocks level: L5 | Required quality requirement: full

Evidence: Coverage near 60%, adversarial tests missing.

Action: Reach at least 80% source code coverage with integration and adversarial tests.

Nice to Have

No nice-to-have gaps.

Gap Summary

SUB-CHARACTERISTIC	CATEGORY	GAP	BLOCKS LEVEL	PRIORITY
Fairness	responsibility	small	L1	critical
Responsiveness	utility	large	L1	critical

SUB-CHARACTERISTIC	CATEGORY	GAP	BLOCKS LEVEL	PRIORITY
Vulnerability	responsibility	small	L1	critical
Accuracy	utility	small	L3	important
Discoverability	comprehensibility	large	L3	important
Effectiveness	utility	large	L3	important
Operability	modifiability	small	L3	important
Understandability	comprehensibility	small	L3	important
Adaptability	robustness	small	L4	important
Explainability	responsibility	small	L4	important
Repeatability	productionizability	small	L4	important
Traceability	comprehensibility	small	L4	important
Cost-Effectiveness	economy	large	L5	important
Efficiency	economy	small	L5	important
Maintainability	modifiability	small	L5	important
Modularity	modifiability	small	L5	important
Monitoring	productionizability	small	L5	important
Readability	comprehensibility	small	L5	important
Resilience	robustness	small	L5	important
Scalability	robustness	large	L5	important
Testability	modifiability	small	L5	important

Method: gap scores no=0, small=1, large=2; quality score uses denominator 2 x assessed sub-characteristics. Maturity level computed from Table 1 level gates encoded in configs.

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